

Laidlaw Progress Brief — Week 2

From Meeting Recalibration to a Manuscript-Ready Pathway Panel

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*Purpose: show what Week 2 changed after the 17 July lab meeting,
what is now real versus provisional, and what unlocks inference.*

No Hospital Authority outcome coefficients are claimed.

1. Where things stand

Week 1 ended with a clear boundary: HKO climate descriptives were real; models were synthetic; HA outcomes were pending. The 17 July meeting then **recalibrated the scientific target**: the general HA file does not specify reasons for admission, so AMI/principal-dx CVD is out of scope from that extract; near-term work uses **stroke admission aggregates**; and we should explore a labelled panel of ~10 methods rather than one hero specification.

Central point this week: the public data layers and the analysis system are now ready for that recalibrated question. Real EPD pollution and CHP flu are in; a 17-pathway panel dry-runs cleanly; stroke aggregates remain the single missing keystone.

2. Scientific framing (updated)

*Monthly thermal extremes and stroke admission aggregates in Hong Kong, 2013–2023:
a multi-specification analysis complementary to daily heatwave–mortality work.*

We estimate ecological monthly burden associations — not daily DLNM triggering, and not excess-death counts from the lab’s multi-definition heatwave–mortality line. Cold remains co-primary (Goggins; Yang/Chong 2025). Heatwave definition diversity is intentional (Guo/Gasparrini 2017; Wang/Ren 2019; Liu/Ren/Bishai).

3. What became real this week

Pollution (EPD EPIC). Official monthly general-station means for NO₂, O₃, PM_{2.5}, and PM₁₀ (2013–2023). Early (2013–2016) to late (2021–2023): NO₂ –32%, PM_{2.5} –45%, PM₁₀ –38%; O₃ +32%. Staged into pathway P11 (O₃ last).

Influenza (CHP Flu Express). Monthly positivity for pathway P14; coverage 121/132 months (early-2013 gap documented; complete-case).

Analysis exposures. Lag-1 temperature (Dec 2012 joined), multi-percentile heatwave-month indicators (p90/p95/p97.5), daily diurnal-range temperature variability, Ren/Wang spell and 2D3N metrics — all in `exposures_monthly_2013_2023.csv`.

Climate reminder (unchanged ground truth). Hot nights and very hot days rose sharply after 2018; cold days persist. Peak hot-night year: 2021 (61 nights). These remain **descriptive climate findings only**.

4. What I built (workflow, not findings)

- Executable pathway registry P01–P18 with Statistical Analysis Protocol, scientist runbook, and stroke-aggregate schema.
- End-to-end orchestrator: `Rscript scripts/run_pathway_pipeline.R` (dev) or `PATHWAY_MODE=real` after placing aggregates.
- Dry-run on synthetic stroke with **real** climate/pollution/flu: **17 pathways OK**; P13 disabled until subtype exists.
- Manuscript products: coefficient panel, headline diagnostics, forest figure, table shells.
- Headline proposal for Gate 3: **P02** (Tmax/Tmin) + **P04** (official extreme-day counts).

Synthetic coefficients are not study results.

5. Still missing / recommended defaults

- **Missing:** approved stroke aggregate file + dictionary; PI governance path for use; Gate 3 freeze after real descriptives.
- **Defaults:** monthly stroke aggregates (pooled unless IS/HS present); age×sex grain if available; negative binomial with $\log(\text{population} \times \text{days})$ offset; panel reporting, not seventeen discoveries; no AMI claims from files without admission reasons.

6. Ask

When aggregates clear transfer, place the CSV under `data_raw/ha_secure_placeholder/` and run real mode. First deliverable: QC receipt + pathway panel + forest, then a short Gate 3 freeze note naming the headline pair.

Deck: `reports/Week2.Progress Deck.pdf` *Repo:* <https://github.com/bobshenruililin/Laidlaw-Heat-Project>